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End Semester Examination: 2024

Course: B.Tech. Mechanical Engineering/ Aerospace Engineering

Semester: 3rd

Paper Name: Engineering Mathematics-III

Subject Code: TMA - 303

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question.
- (iii) Total marks in each main question are twenty.
- (iv) Each sub question carries 10 marks.

Q 1

20 Marks

- (a) Show that the given function $u(x, y) = \frac{1}{2} \log(x^2 + y^2)$ is harmonic. Also find its harmonic conjugate function.

CO1

- (b) Using Milne's Thomson method to find an analytic function $f(z)$, where the real part of function $f(z) = u(x, y) + iv(x, y)$ is $u(x, y) = e^x \cos y$.

CO1

- (c) Show that the function $f(z) = u + iv$, where $f(z) = \begin{cases} \frac{x^2 y^5 (x + iy)}{x^4 + y^{10}}, & z \neq 0 \\ 0, & z = 0 \end{cases}$

is not analytic at $z = 0$ although Cauchy-Riemann equations are satisfied at $z = 0$.

CO1

Q 2

20 Marks

- (a) Using Cauchy's integral formula to compute the value of the integral

$$\oint_C \frac{\sin \pi z^2 + \cos \pi z^2}{(z-1)(z-2)} dz, \text{ where } C: |z| = 3.$$

CO2

- (b) Write down the statement of Laurent's Theorem. Find the Laurent's series expansion

of the function $f(z) = \frac{z^2 - 1}{z^2 + 5z + 6}$ in the domain $2 < |z| < 3$.

CO2

- (c) Define the singularity of a complex valued function $f(z)$, and discuss the nature of singularity of $f(z) = \frac{\sin z}{z}$ about $z = 0$.

Q 3

20 Marks

- (a) Using Bisection method, to find a real root of an equation $x \log_{10} x = 1.2$ correct to three decimals.

CO3

- (b) A train is moving at the speed of 30 m/sec. Suddenly brakes are applied. The speed of the train per second after ' t ' seconds is given below:

Time (t)	0	5	10	15	20	25	30	35	40	45
Speed (v)	30	24	19	16	13	11	10	8	7	5

Using Simpson's $\frac{3}{8}$ rule to compute the distance moved by the train in 45 seconds.

CO3

- (c) What is an errors? Given $x = 0.005998$, if x is rounded off to three decimal places, then compute the absolute, relative and percentage errors

CO3

Q 4

20 Marks

- (a) Solve $\frac{dy}{dx} = xy$ numerically from $x = 0$ to $x = 1$ using Euler method with initial condition of $y(0) = 1$ and step size (h) of 0.2. Find the value of $y(1)$. (Correct to three decimal places).

CO5

- (b) Using Newton-Raphson method to compute a root of the equation $x^3 - x - 1 = 0$ with initial condition $x_0 = 1.6$, (Correct to three decimal places).

CO3

- (c) Given that the initial value problem $\frac{dy}{dx} = x + y$ with initial condition $y(0) = 1$. Using Runge- Kutta method to obtain the value of $y(0.1)$ correct to three decimal places, (Taking step size $h = 0.1$).

CO5

Q 5

20 Marks

(a) If the probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} kx^2; & 0 < x < 1 \\ 0; & \text{otherwise} \end{cases}$$

Find the value of k and hence compute its mean, variance, and standard deviation.

CO6

(b) The following are the values of two variables, diastolic blood pressure (y) and time in minutes (x). Diastolic blood pressure (y) was recorded at various time periods (x) on individuals. Find the regression line of y on x and x on y .

Time periods (x)	0	5	10	15	20
Diastolic blood pressure (y)	72	67	70	65	66

CO6

(c) The number of airplane landings at an airport in a minute (X) and their probabilities are given by

X_i	0	1	2	3	4	5	6
$p(x_i)$	0.02	0.15	0.22	0.26	0.17	0.14	0.04

Find the mean and standard deviation of X .

CO6

